

Name

Index Number

THE WEST AFRICAN EXAMINATIONS COUNCIL

West African Senior School Certificate Examination
for School Candidates

SC 2024

INTEGRATED SCIENCE 1
[50 marks]

1 hour

Do not open this booklet until you are told to do so. While you are waiting, read and observe the following instructions. Write your name and index number in the spaces provided above.

- Use 2B pencil throughout.
- On the pre-printed answer sheet, check that the following details are correctly printed:
 - In the space marked **Name**, check your **surname** followed by your **other names**.
 - In the spaces marked *Examination*, *Year*, *Subject* and *Paper*, check 'WASSCE', 'SC 2024', 'INTEGRATED SCIENCE', and '1' respectively.
 - In the box marked *Index Number*, your **index number** has been printed vertically in the spaces on the left-hand side, and each numbered space has been shaded in line with each digit. **Reshade** each of the shaded spaces.
 - In the box marked *Subject Code*, the digits 517113 are printed vertically in the spaces on the left-hand side. **Reshade** the corresponding numbered spaces as you did for your index number.
- An example is given below. This is for a male candidate whose name is Edem Koki MUSAH. His *index number* is 7102143958 and he is offering *Integrated Science 1*.

THE WEST AFRICAN EXAMINATIONS COUNCIL
ANSWER SHEET

PRINT IN BLOCK LETTERS	
Name: MUSAH EDEM KOFI	GHA
Examination: WASSCE	Year: SC 2024
Subject: INTEGRATED SCIENCE	Paper: 1

INSTRUCTIONS TO CANDIDATES	
1. Use grade 2B pencil throughout.	
2. Answer each question by choosing one letter and shading like this: A ☐ B ☐ C ☐ D ☐	
3. Erase completely any answer you wish to change.	
4. Leave extra spaces blank if the answer spaces provided are more than you need.	
5. Do not make any markings across the heavy black marks at the right hand edge of your answer sheet.	

INDEX NUMBER										SUBJECT CODE									
7	☐	☐	☐	☐	☐	☐	☐	☐	☐	5	☐	☐	☐	☐	☐	☐	☐	☐	☐
1	☐	☐	☐	☐	☐	☐	☐	☐	☐	1	☐	☐	☐	☐	☐	☐	☐	☐	☐
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2	☐	☐	☐	☐	☐	☐	☐	☐	☐	1	☐	☐	☐	☐	☐	☐	☐	☐	☐
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3	☐	☐	☐	☐	☐	☐	☐	☐	☐	For Supervisors only If candidate is absent shade this space ☐									
9	☐	☐	☐	☐	☐	☐	☐	☐	☐										
5	☐	☐	☐	☐	☐	☐	☐	☐	☐										
8	☐	☐	☐	☐	☐	☐	☐	☐	☐										

Answer **all** the questions.

Each question is followed by four options lettered A to D. Find the **correct** option for **each** question and shade in **pencil** on your answer sheet, the answer space which bears the same letter as the option you have chosen.

Give only **one** answer to **each** question. An example is given below.

Which of the following elements is a metal?

- A. Carbon
- B. Copper
- C. Helium
- D. Krypton

The correct answer is Copper, which is lettered **B**, and therefore answer space **B** would be shaded.

☐ A ☒ B ☐ C ☐ D

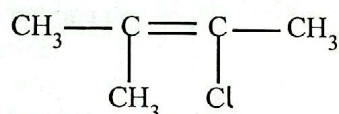
Think carefully before you shade the answer spaces; erase completely any answers you wish to change.

Do all rough work on this question paper.

Now answer the following questions.

1. Organic compounds are **mostly**
 - A. soluble in water.
 - B. not inflammable.
 - C. covalent.
 - D. ionic.
2. Fifteen children from a class of 45 roll tongues. What percentage of the class are rollers?
 - A. 25.3 %
 - B. 33.3 %
 - C. 35.3 %
 - D. 75.3 %
3. An electric bulb is rated 120 W and 240 V. Determine the current it draws from the mains.
 - A. 0.5 A
 - B. 0.6 A
 - C. 1.0 A
 - D. 2.0 A
4. Which of the following organisms has **neither** a nucleus nor membrane bound organelles?
 - A. Bacteria
 - B. Mucor
 - C. Paramecium
 - D. Spirogyra

5. Which of the following compounds is a hydrocarbon?
- C_5H_8
 - $C_6H_{12}O_6$
 - CH_3CH_2OH
 - CH_3COCH_3
6. Turgidity in plants is the result of
- water entering the cells.
 - water leaving the cells.
 - the withdrawal of cytoplasm from the cell wall.
 - diffusion of materials into the cells.
7. Mineral salts from the soil pass into the root hair cells by
- turgor pressure.
 - active transport.
 - osmosis.
 - capillarity.
8. For grafting to be successful, the scion and stock **must** be of the same
- size.
 - length.
 - diameter.
 - texture.
9. The IUPAC name of the compound whose structural formula is illustrated below is



- 2-chloro-3-methylbut-2-ene.
 - 2-methyl-2-chlorobut-2-ene.
 - 2-chloro-3-methyl-2-butane.
 - 2-methyl-2-chloro-2-butyne.
10. An example of a wind instrument is
- flute.
 - guitar.
 - violin.
 - xylophone.
11. Fluid consists of
- solid and liquid only.
 - liquid and gas only.
 - gas and solid only.
 - liquid, solid and gas.
12. Living organisms which help to modify the physical and chemical composition of the soil are
- earthworms.
 - lichens.
 - bacteria.
- I and II only.
 - I and III only.
 - II and III only.
 - I, II and III.

13. Cowpea plants are used in crop rotation system of farming so that
- grazing animals would not destroy the crops.
 - nitrates would be added to the soil.
 - soil erosion could be checked.
 - weeds would be suppressed.
14. A dibber is used for
- dehusking.
 - pruning.
 - thinning.
 - planting.
15. Which of the following compounds is a neutral oxide?
- CO_2
 - NO_2
 - MgO
 - CO
16. Energy output in a device is equal to work done
- by the load.
 - on the load.
 - by the effort.
 - on the effort.
17. The **main** function of ileum in mammals is
- digestion and assimilation.
 - assimilation and absorption.
 - digestion and absorption.
 - ingestion and digestion.
18. An example of inorganic fertilizer is
- compost.
 - farmyard manure.
 - green manure.
 - urea.
19. Proteins are **usually** made up of
- sulphur, carbon and nitrogen.
 - carbon, hydrogen and nitrogen.
 - carbon, hydrogen and oxygen.
 - oxygen, carbon and sulphur.
20. Which of the following statements about sea breeze are **correct**? Sea breeze occurs
- during the day.
 - during the night.
 - as a result of convection currents.
- I and II only
 - I and III only
 - II and III only
 - I, II and III

21. Examples of plants with rhizome are
- ginger and fern.
 - onion and ginger.
 - yam and Irish potato.
 - onion and ferns.
22. An agricultural system in which crops and animals are produced on the same piece of land at the same time is termed
- continuous cropping.
 - mixed cropping.
 - mixed farming.
 - pastoral farming.
23. Which of the following non-metals conducts electricity?
- Diamond
 - Graphite
 - Silicon
 - Vanadium
24. A characteristic feature of metamorphic and sedimentary rocks is that they are
- crystalline in structure.
 - non-crystalline in structure.
 - glassy in appearance.
 - metallic in appearance.
25. The management practice which makes the transport of farm animals easy is
- castration.
 - dehorning.
 - identification.
 - quarantine.
26. Which of the following statements about compounds are **correct**?
- They are products of chemical combination.
 - Their constituents are present in fixed ratio by mass.
 - Their constituents keep their individual properties.
- I and II only
 - I and III only
 - II and III only
 - I, II and III
27. The number of electrons present in the outermost shell of an element with atomic number 20 is
- 2
 - 6
 - 8
 - 10
28. Which of the following salts can cause temporary hardness in water?
- $\text{Mg}(\text{HCO}_3)_2$
 - $\text{Fe}_2(\text{SO}_4)_3$
 - MgCl_2
 - CaSO_4

29. Which of the following components of the electromagnetic spectrum has the **highest** frequency?
- Gamma rays
 - Infra-red radiations
 - Microwave
 - Ultraviolet radiations
30. In mammals, the organ which deaminates excess amino acid to form urea is the
- kidney.
 - liver.
 - pancreas.
 - spleen.
31. The equivalent of 161 K in degree celcius is
- 161 ° C
 - 112 ° C
 - 112 ° C
 - 161 ° C
32. The shape of the lens of the human eye is varied by the action of the
- iris.
 - pupil.
 - ciliary muscle.
 - suspensory ligament.
33. Maintenance ration provides farm animals with nutrients required for
- egg production.
 - vital body processes.
 - growth and development.
 - development and maturity.
34. The set-up used to separate water from ethanol should include
- separating funnel.
 - burette.
 - condenser.
 - volumetric flask.
35. Which of the following organelles is the site for protein synthesis?
- Golgi body
 - Mitochondrion
 - Nucleus
 - Ribosome
36. Fehling's solution is used in the laboratory to test for the presence of
- fats.
 - starch.
 - reducing sugars.
 - fatty acids.

37. The process by which liquid changes into vapour at a particular temperature is known as
- boiling.
 - condensation.
 - evaporation.
 - melting.
38. Which of the following quantities have **both** magnitude and direction?
- Displacement
 - Speed
 - Momentum
- I and II only
 - I and III only
 - II and III only
 - I, II and III
39. Two resistors **each** $5\ \Omega$ are connected in parallel. Determine their effective resistance.
- $2.5\ \Omega$
 - $6.4\ \Omega$
 - $10.0\ \Omega$
 - $25.0\ \Omega$
40. A gas that gives a pop sound with a flame is
- ammonia.
 - carbon (IV) oxide.
 - hydrogen.
 - nitrogen.
41. The neutrons and protons in the nucleus of an atom are collectively called
- mass number.
 - neutrons.
 - nucleons.
 - atomic number.
42. An example of naturally occurring weak acid is
- ammonia.
 - citric acid.
 - hydrochloric acid.
 - nitric acid.
43. A person who suffers from myopia
- cannot see at all.
 - can see clearly from far.
 - can see clearly objects closer.
 - wears a converging lens for correction.
44. Which of the following characteristics are **correct** about dairy cattle? They have
- large udder,
 - long legs,
 - wedged body.
- I and II only
 - I and III only
 - II and III only
 - I, II and III

45. The layer of the atmosphere in which air travel is most convenient is
- mesosphere.
 - stratosphere.
 - thermosphere.
 - troposphere.
46. Breeds of fowl which are reared for the purpose of producing meat are called
- broilers.
 - chicks.
 - hens.
 - layers.
47. Which of the following processes uses solar energy?
- Drying materials,
 - Generation of electricity,
 - Heating of food substances.
- I and II only
 - I and III only
 - II and III only
 - I, II and III
48. The angle of reflection of a ray is 40° . Determine the angle between the surface of the mirror and the ray.
- 40°
 - 50°
 - 80°
 - 180°
49. The **major** cause of post-harvest loss of stored millet in Ghana is
- bird attack.
 - insect attack.
 - mechanical injury.
 - high humidity.
50. Group of organs which form part of the central nervous system are the
- brain, the eyes and the ears.
 - brain and the spinal cord.
 - brain, the ears and the spinal cord.
 - eyes and the spinal cord.

END OF PAPER

SC5172
WASSCE 2024
INTEGRATED
SCIENCE 2
Essay Test
1½ hours

2

CANDIDATE'S NAME	
INDEX NUMBER	SIGNATURE
DATE:	

THE WEST AFRICAN EXAMINATIONS COUNCIL

West African Senior School Certificate Examination
for School Candidates

SC 2024

INTEGRATED SCIENCE 2
[Essay]

1½ hours

INSTRUCTIONS TO CANDIDATES

- In the spaces provided above, insert your name, full index number, normal signature and the date of examination.*
- This booklet consists of six questions. Answer four questions in all. All questions carry equal marks.*
- Write the question number and your index number at the top of each page.*
- Write on both sides of the paper unless otherwise instructed on the question paper.*
- Begin each answer to a question on a fresh page. Leave two lines between answers where there are sub-sections to the same question.*
- On no account should you tear off any part of the booklet. It is an examination malpractice to do so. The entire booklet will be collected at the end of the examination.*
- Write in the space provided below, the number of the questions you have answered in the order in which you have written them.*

For Examiner's Use Only	
Question Number	Mark
TOTAL	

2
PAPER 2
ESSAY
[80 marks]

1½ hours

Answer four questions only from this section.

Credit will be given for clarity of expression and orderly presentation of material.

All questions carry equal marks.

1. (a) (i) What are *chromosomes*? [2 marks]
(ii) State **three** uses of DNA testing in humans. [3 marks]
- (b) (i) Name **three** noble gases present in the air. [3 marks]
(ii) Explain **briefly** why the noble gases are unreactive under normal conditions. [2 marks]
- (c) (i) Explain the term *post harvest loss* as used in crop production. [2 marks]
(ii) State **three** precautions farmers take to minimize post harvest losses of cereals in Ghana. [3 marks]
- (d) (i) What is an *inductor* as used in electronics? [2 marks]
(ii) State **three** applications of solar energy in everyday life. [3 marks]
2. (a) Give **two** examples of **each** of the following types of crops cultivated in Ghana:
(i) stem tubers; [6 marks]
(ii) root tubers; [2 marks]
(iii) legumes. [2 marks]
- (b) (i) Explain the term *saponification*. [2 marks]
(ii) State the observation that could be made when brine is added to the products of saponification. [2 marks]
- (c) (i) What is a *reflex action*? [2 marks]
(ii) Give **three** examples of reflex actions in the human body. [3 marks]
- (d) (i) State the law of *conservation of energy*. [2 marks]
(ii) A cement block of mass 25 kg is placed at a height of 10 m above the ground and pushed to fall. Calculate the velocity of the block just before it touches the ground. [3 marks]
 $[g = 10 \text{ m s}^{-2}]$
3. (a) (i) State **three** causes of genetic variation in human population. [3 marks]
(ii) Give **two** ways in which genetic variation affects organisms. [2 marks]
- (b) (i) Explain the following terms associated with air movement:
(a) Trade winds;
(b) Storm. [4 marks]
- (ii) A storm moves at a speed of $2.0 \times 10^2 \text{ m s}^{-1}$. It blows for 30 minutes at this constant speed. How far would the storm reach. [2 marks]
- (c) (i) What are *radioisotopes*? [2 marks]
(ii) State **three** uses of radioisotopes. [3 marks]

- (d) State **four** signs that a cow exhibits just before parturition. [4 marks]
4. (a) (i) What is an *organic functional group*? [2 marks]
 (ii) Name the functional group of **each** of the following organic compounds:
 (α) HCOOH ;
 (β) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$;
 (γ) CH_3CHCH_2 . [3 marks]
- (b) (i) State **two** methods farmers adopt to control rodents on the farm. [2 marks]
 (ii) Give **one** way in which **each** of the following factors is of importance when selecting a site for lettuce cultivation:
 (α) type of soil;
 (β) accessibility to site;
 (γ) topography. [3 marks]
- (c) (i) What is *personal hygiene*? [2 marks]
 (ii) State **three** effects of poor personal hygiene on the individual. [3 marks]
- (d) (i) What is an *electromagnet*? [2 marks]
 (ii) State **three** uses of electromagnets in everyday life. [3 marks]
5. (a) (i) Explain the term *brooder house* as used in poultry. [2 marks]
 (ii) State **three** causes of cannibalism in poultry. [3 marks]
- (b) (i) What is a *rainbow*? [2 marks]
 (ii) Explain **briefly** how a rainbow is formed. [3 marks]
- (c) (i) Name **two** types of interactions that exist in an ecosystem. [2 marks]
 (ii) State **three** ways by which energy could be lost in food chain. [3 marks]
- (d) (i) What is a *solvent*? [2 marks]
 (ii) A solution was prepared by dissolving 20 g of NaOH in 500 cm^3 of distilled water. Calculate the concentration of the solution in mol/dm^3 .
 [Na = 23.0, O = 16.0, H = 1.0] [3 marks]
6. (a) (i) Explain **briefly** how static electricity is produced. [2 marks]
 (ii) Name **two** other sources of electric power generation apart from hydro and thermal. [2 marks]
- (b) (i) Name **three** glands associated with the alimentary canal of humans. [3 marks]
 (ii) List the substance secreted by **each** of the glands named in (b)(i). [3 marks]
- (c) (i) State **one** use of **each** of the following gases:
 (α) hydrogen;
 (β) ammonia. [2 marks]
- (ii) Write a balanced chemical equation for the reaction involving ammonium chloride and calcium hydroxide. [3 marks]

(d) Name the **main** nutrient that is responsible for each of the following processes in crop plants:

- (i) root and tuber development;
- (ii) formation of chlorophyll;
- (iii) development of strong stem;
- (iv) growth;
- (v) fruit development.

[5 marks]

END OF PAPER

SC5173
WASSCE 2024
INTEGRATED
SCIENCE 3
Test of Practical Work
2 hours

3

CANDIDATE'S NAME

INDEX NUMBER

SIGNATURE

DATE:

THE WEST AFRICAN EXAMINATIONS COUNCIL

West African Senior School Certificate Examination
for School Candidates

SC 2024

INTEGRATED SCIENCE 3
- TEST OF PRACTICAL WORK
[60 marks]

2 hours

DIRECTIONS TO CANDIDATES

1. Do **not** turn over this booklet until you are told to do so.
2. In the spaces provided above, insert your **name**, **full Index number**, **normal signature** and the **date of examination**.
3. Answer **all** the questions in this booklet.
4. On **no account** should you tear **any** part of the booklet. The entire booklet will be collected at the end of the test.
5. **All rough work should be done in this booklet** and then crossed out neatly. Under **no circumstance** should you work on **any** other paper.
6. Write the **NUMBERS OF THE QUESTIONS YOU HAVE ANSWERED** in the order in which you have answered them, in the space provided below.

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For Examiner's
Use Only

Question Number	Mark
TOTAL	

- 2
1. Fig. 1 is an illustration of two methods of vegetative propagation in flowering plants labelled A and B at various stages A₁, A₂, A₃, A₄, B₁ and B₂. Study the figure carefully and answer the questions that follow.

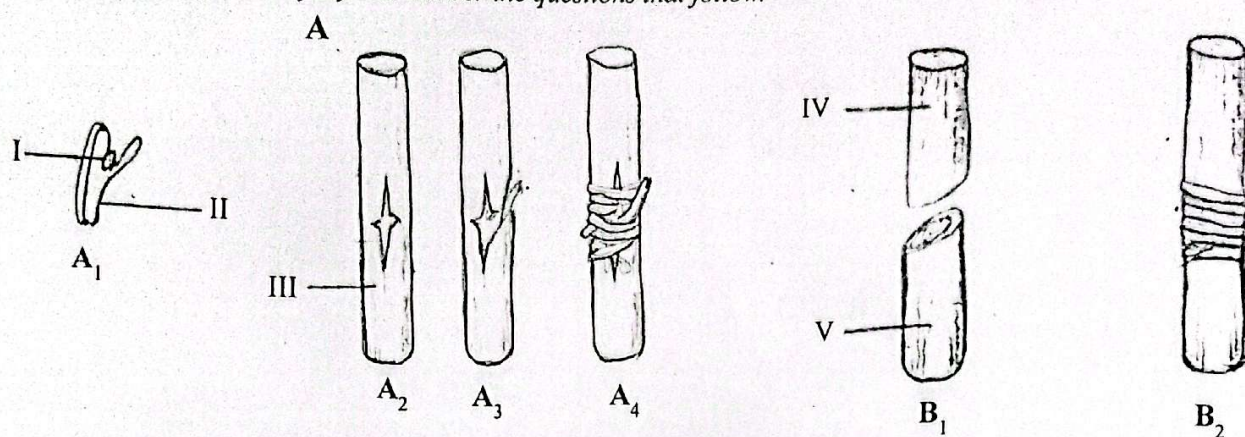


Fig. 1

- (a) Name **each** of the methods of propagation illustrated in A and B.
- A.....
- B.....
- [2 marks]
- (b) Name **each** of the parts labelled I, II, III, IV and V.
- I.....
- II.....
- III.....
- IV.....
- V.....
- [5 marks]
- (c) State **one** observable feature **each** for the stages A₁, A₂, A₃, B₁ and B₂.
- A₁.....
- A₂.....
- A₃.....
- B₁.....
- B₂.....
- [5 marks]
- (d) Stage **two** precautions to be taken in the method labelled A.
-
-

[2 marks]

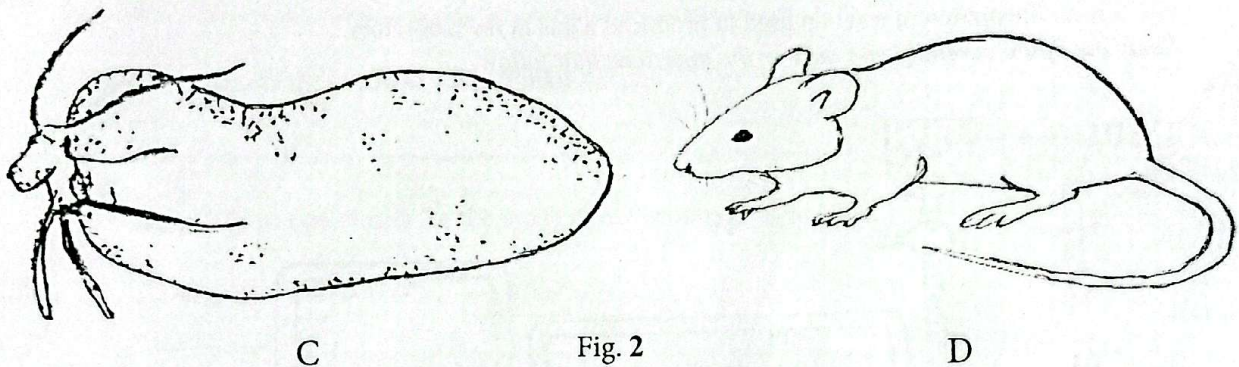
- (e) Give **one** local example **each** of plants that could be propagated by the method illustrated in:

(i) A:

(ii) B:

[2 marks]

2. Fig. 2 is an illustration of a field crop labelled C and a crop pest labelled D.
Study the figure carefully and answer the questions that follow.



- (a) Name the group of crops to which the crop labelled C belongs.

[1 mark]

- (b) Give **one** characteristic feature of the crop labelled C.

[1 mark]

- (c) Describe **briefly** how crop C could be propagated.

[3 marks]

- (d) State **three** cultural practices that could promote the healthy growth of the crop.

[3 marks]

- (e) (i) State **one** kind of damage caused by pest D to crop C.

[1 mark]

- (ii) Name the group of pests to which pest D belongs.

[1 mark]

- (iii) Name **two** other organisms that belong to the group named in (e)(ii).

[2 marks]

- (f) State **two** ways in which pest D could be controlled on a farm of crop C.

[2 marks]

3. Fig. 3 is an illustration of a set up used to produced a gas in the laboratory.
Study the figure carefully and answer the questions that follow.

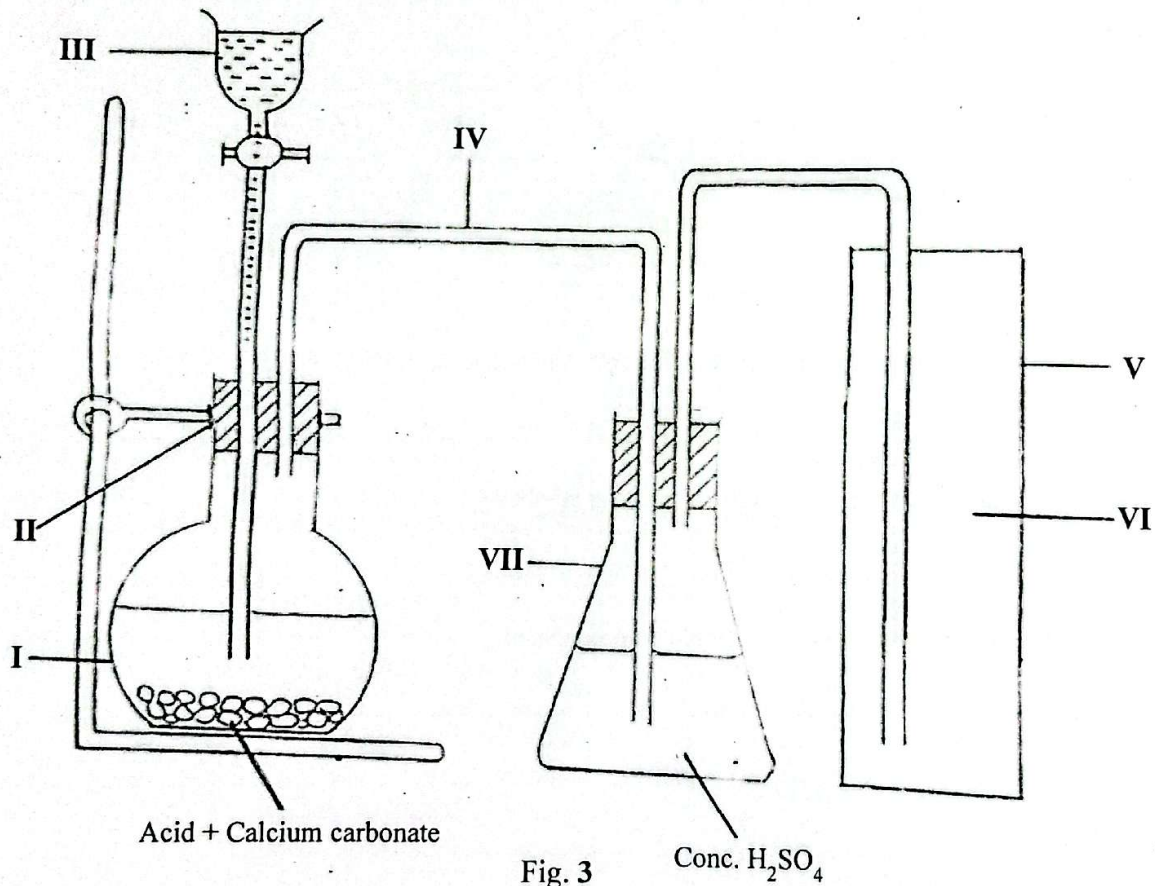


Fig. 3

- (a) Name **each** of the parts labelled I, II, III, IV and V.

I.....

II.....

III.....

IV.....

V.....

[5 marks]

(b) State the function of **each** of the parts labelled:

(i) IV;.....

(ii) Conc. H_2SO_4 [2 marks]

(c) Name **two** suitable acids that could be used in this experiment.

.....

..... [2 marks]

(d) Name the gas produced from the experiment.

..... [1 mark]

(e) State **one** physical property of the gas produced.

..... [1 mark]

(f) (i) State the mode of collection of the gas.

..... [1 mark]

(ii) Give **one** reason for the mode of collection stated in (f)(i).

..... [1 mark]

(g) Describe **briefly** the test for the gas produced.

.....

.....

.....

..... [2 marks]

4. Fig. 4 is an illustration of an experiment to investigate the relationship between the power P of an electrical appliance **M** and current I flowing through the appliance.

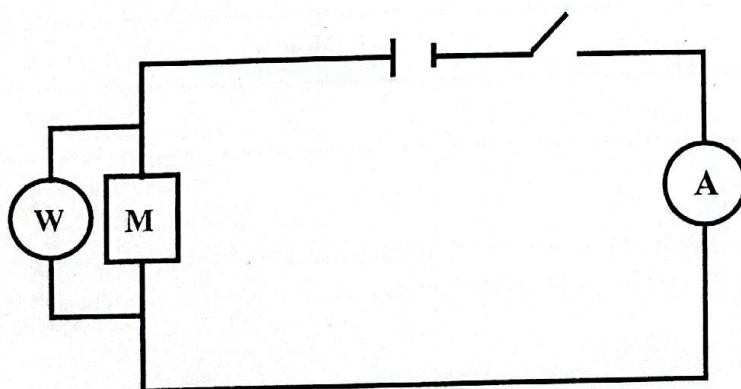


Fig. 4

M represents the electrical appliance and W the wattmeter which measures the electrical power. The current through the appliance was varied five times and each time the power (as indicated by the wattmeter) was measured. Fig. 4(a) shows the current $I = I_1, I_2, I_3, I_4$ and I_5 . Fig. 4(b) shows the power $P = P_1, P_2, P_3, P_4$ and P_5 .

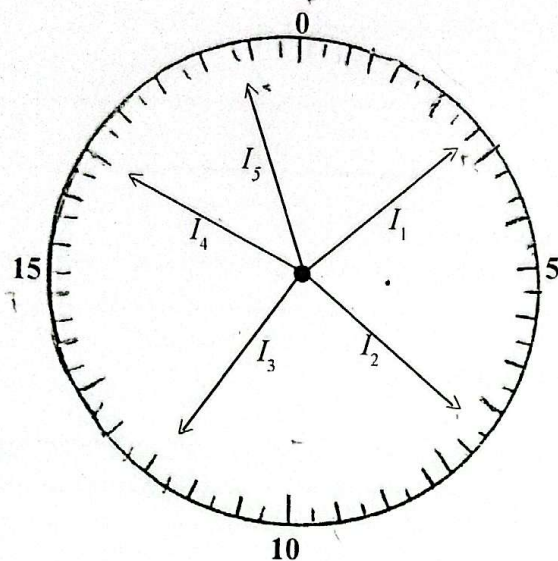


Fig. 4(a)

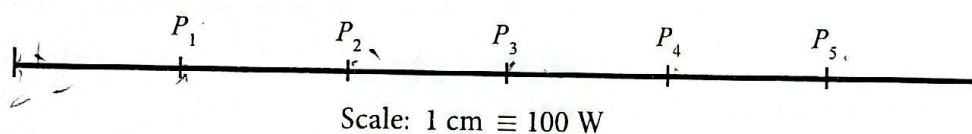


Fig. 4(b)

- Read and record the values of current $I = I_1, I_2, I_3, I_4$ and I_5 .
- Read and record the values of $P = P_1, P_2, P_3, P_4$ and P_5 .
- Convert the raw values of power P into actual values of power P using the scale provided.
- Copy and complete the table as shown below:

Values of Current / A	$I_1 =$	$I_2 =$	$I_3 =$	$I_4 =$	$I_5 =$
Raw values of Power / cm	$P_1 =$	$P_2 =$	$P_3 =$	$P_4 =$	$P_5 =$
Actual values of Power / W	$P_1 =$	$P_2 =$	$P_3 =$	$P_4 =$	$P_5 =$

- Plot a graph of actual values of power P on the vertical axis and the current I on the horizontal axis using the graph sheet provided on page 8.

[7 marks]

[5 marks]

(f) Determine the slope of the graph.

.....

.....

.....

.....

[2 marks]

(g) State the significance of the slope.

.....

.....

.....

[1 mark]

END OF PAPER